

A decorative pattern of green squares and rectangles of varying shades, arranged in a grid-like fashion that tapers off towards the right side of the image. The pattern is set against a dark gray background.

SplitsFromVideo

Milestone 1

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Comparing tools

- Computer vision (OpenCV)
 - standard, many features
- Optical Character Recognition (PaddleOCR)
 - High accuracy with numbers
- YouTube navigation/streaming
 - yt-dlp for video streams
 - cap-from-youtube for getting video frames





Demos

cap-from-youtube

```
# 2. Open stream with OpenCV
cap = cv2.VideoCapture(video_url)

# 3. Seek and capture frame 300
target_frame = 300
cap.set(cv2.CAP_PROP_POS_FRAMES, target_frame)
```



frame_300.jpg



yt-dlp

Separate video and audio stream URLs:

Video (398): <https://rr4---sn-hp57ynl6.googlevideo.com/vid>

Audio (251): <https://rr4---sn-hp57ynl6.googlevideo.com/vid>

```
def get_stream_url(url=DEFAULT_URL): 1 usage 2 ctabin10
    with yt_dlp.YoutubeDL(ydl_opts) as ydl:
        # Don't download
        info = ydl.extract_info(url, download=False)
```



Resolving challenges

- Tool experimentation
 - yt-dlp and cap-from-youtube demos
 - setting up Python environment
 - Minimize on-screen data needed
 - experiment with target resolution
 - look into post-processing
- 

Collaboration tools

- Github repo created
- Google Calendar set up
- Google Docs created and organized



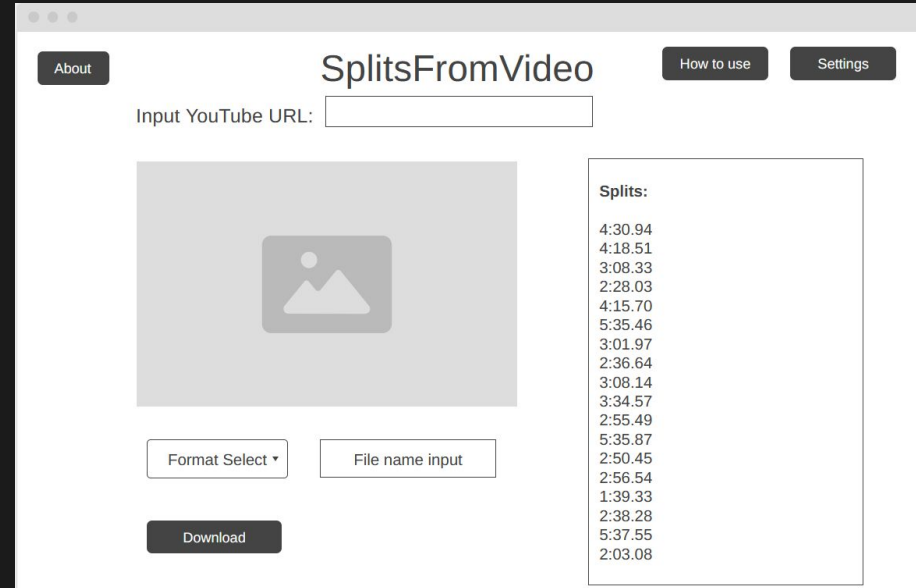
Requirements Document

- Discussed project purpose and scope
- Outlined specific requirements
- Insight into necessary implementation features



Design Document

- Helped us outline classes for the program
- Sketch of user interface



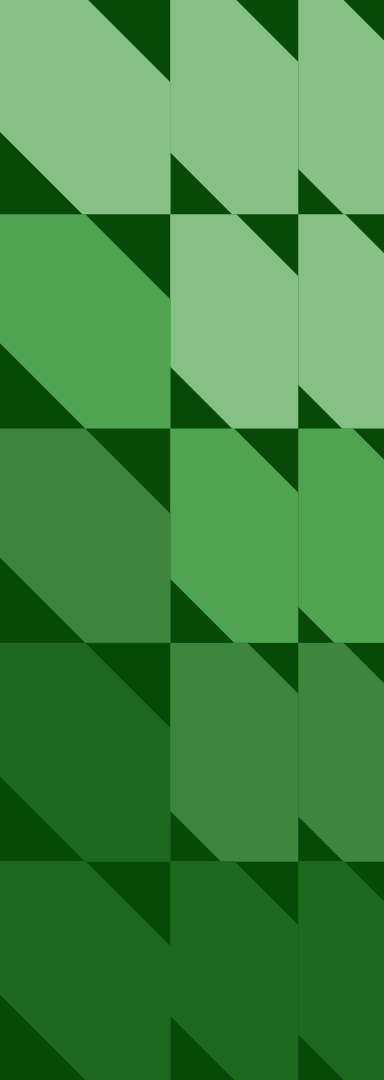
Testing Document

- Outlined scope / out of scope components
- Testing approach & environment
- Test specifications for different components



Next steps (Milestone 2)

- Implement, test, and demo navigating and obtaining specific frames from a YouTube video
- Implement, test, and demo algorithm for finding beginning of splits
- Start shaping user interface in Pywebview



Questions?